

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM watsonx orchestrate + IBM watsonx.ai”

Project Tile – [Career Counseling Companion]

Domain of Project – [Education]

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Problem Statement -

Problem Statement :Agentic Career Counseling Companion

The Challenge

Students often struggle to choose the right career path due to limited guidance, lack of awareness about their strengths and interests, and rapidly changing industry demands. Information about career opportunities, required skills, and future job trends is scattered across multiple sources, making career planning difficult and confusing.

The Objective

Build an **AI-Powered Career Counseling Companion** that provides **personalized career guidance** based on a student's academic background, skills, interests, and goals. The solution should:

Career Recommendations – Suggest suitable career paths.

Profile Analysis – Identify strengths and improvement areas.

Skill Gap Analysis – Highlight missing skills for desired careers.

Learning Roadmap – Generate personalized learning plans.

Certification & Project Suggestions – Recommend relevant certifications and projects.

Career Readiness Assessment – Evaluate preparedness for chosen careers.

Industry Insights – Provide information on emerging opportunities and trends.

Expected Outcome

Help students make informed career decisions, develop relevant skills, and create a clear roadmap for a successful future.

Technology Used

Tech Stack (Mandatory)

Use **any one** of the following approaches:

- IBM watsonx orchestrate + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with LangChain + RAG + IBM Granite Models

I used IBM watsonx orchestrate + IBM watsonx.ai + Granite Models

Proposed Solution -

Proposed Solution – AI-Powered Career Counseling Companion

The proposed solution is an AI-powered **Career Counseling Companion** built using **IBM watsonx Orchestrate, IBM watsonx.ai, and Granite Models** to provide personalized career guidance for students.

The system analyzes a student's academic background, skills, interests, achievements, and career aspirations to recommend suitable career paths. It identifies skill gaps, suggests certifications and projects, and generates personalized learning roadmaps to help students achieve their career goals.

The solution provides:

- Student Profile Analysis
- Career Recommendations
- Skill Gap Identification
- Learning Roadmap Generation
- Certification Recommendations
- Career Readiness Assessment
- Industry Trend Insights

Using IBM Granite models, the system delivers personalized, accurate, and context-aware career guidance that helps students make informed decisions about their future.

Orchestrate Component Used -

1. User Input Interface

Collects student details such as academic background, skills, interests, and career goals.

2. IBM watsonx Orchestrate Agent

Analyzes student profiles and manages personalized career guidance.

3. IBM watsonx.ai (Granite Model)

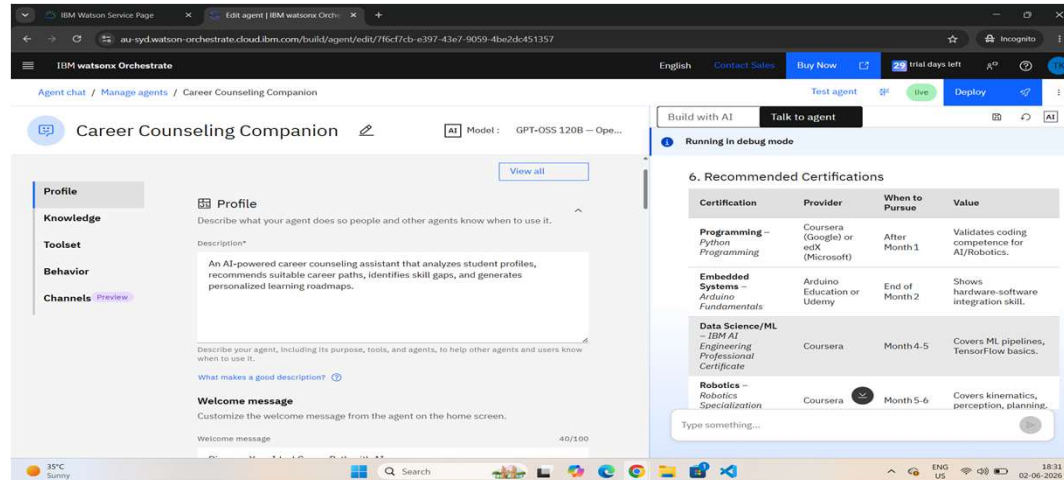
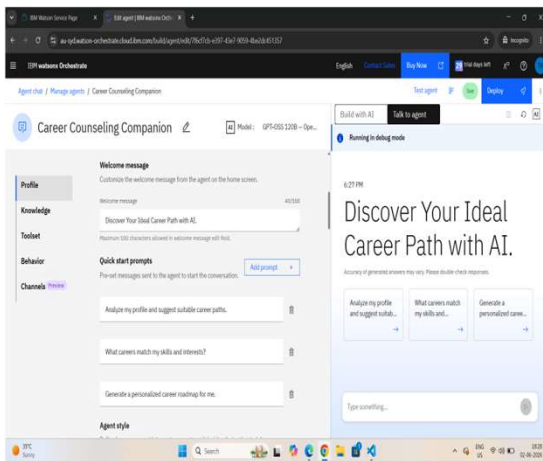
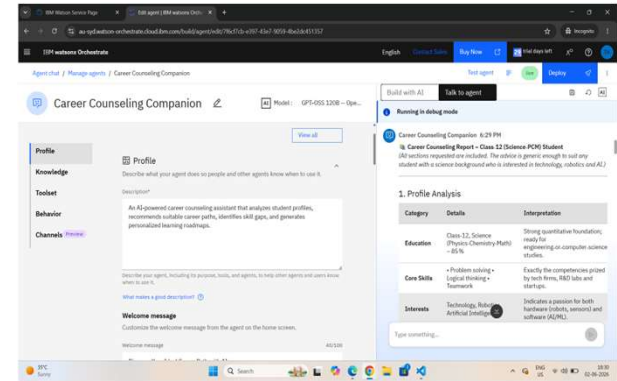
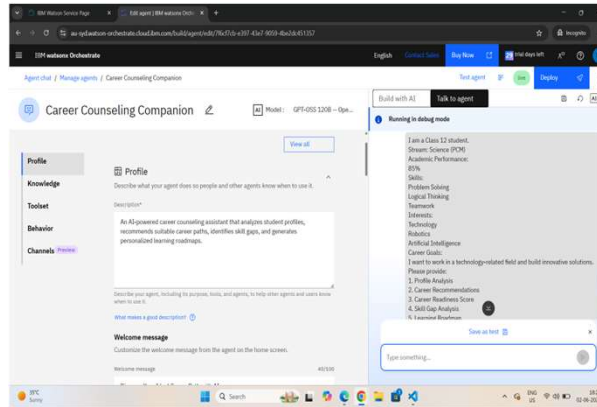
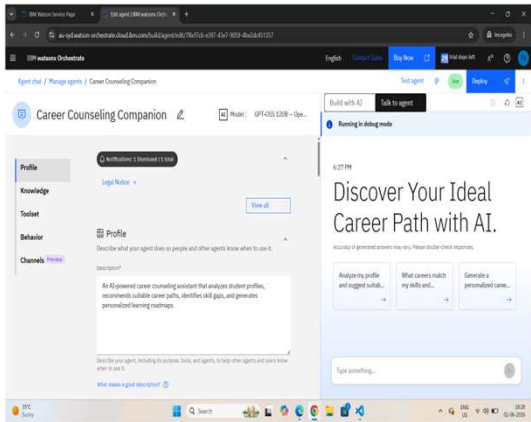
Generates career recommendations, skill-gap analysis, and learning roadmaps.

4. Knowledge Base

Stores career information, certifications, skills, and industry trends.

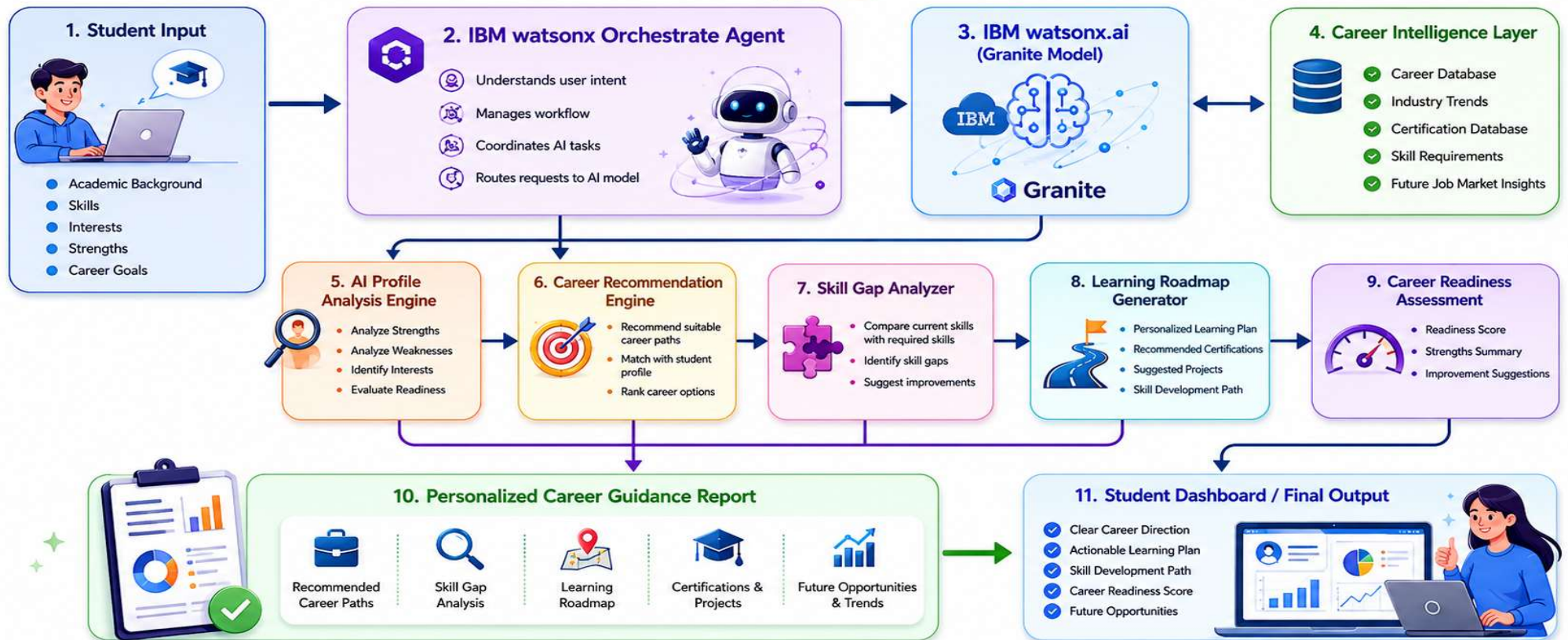
5. Output Interface

Displays personalized career recommendations and future learning plans.



Architecture blueprint

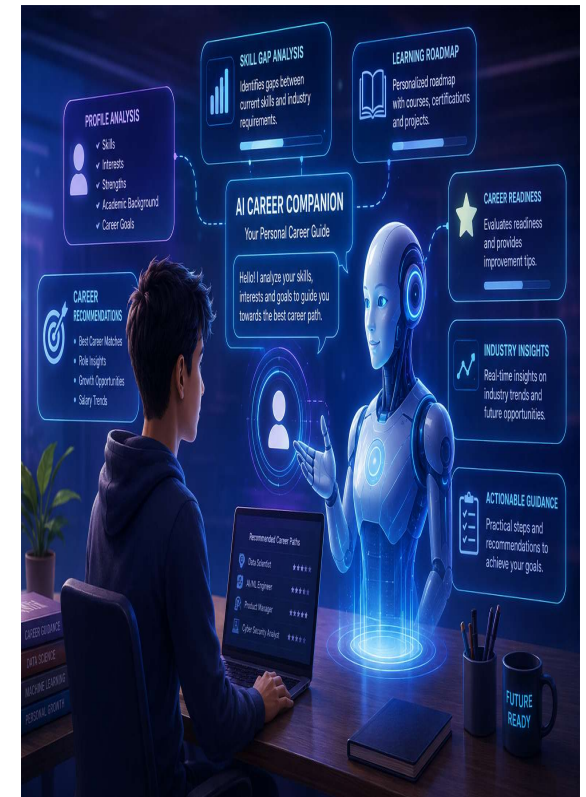
Architecture Workflow – AI-Powered Career Counseling Companion



Role of Agentic AI in the solution

Agentic AI is the core intelligence behind the **Career Counseling Companion**, enabling it to act as a smart and personalized career advisor rather than a simple chatbot. Using **IBM watsonx Orchestrate**, **IBM watsonx.ai**, and **IBM Granite Models**, the system analyzes student profiles based on academic background, skills, interests, and career goals. The AI agent performs **career recommendations**, **skill-gap analysis**, and **career readiness assessment** to identify the most suitable career paths for each student. It also generates personalized learning roadmaps that include relevant certifications, courses, and project suggestions.

Unlike traditional counseling methods, Agentic AI provides **context-aware and adaptive guidance**, continuously aligning recommendations with student aspirations and industry trends. This helps students make informed decisions and prepare for future opportunities. Overall, Agentic AI acts as a **digital career mentor**, empowering students with personalized guidance, actionable insights, and a clear roadmap for achieving their career goals.



Technology Used

- **IBM watsonx Orchestrate** – Agent orchestration and workflow automation
- **IBM watsonx.ai** – AI model execution and intelligent response generation
- **IBM Granite Foundation Models** – Career analysis, reasoning, and recommendation generation
- **IBM Cloud Lite** – Cloud platform for development and deployment
- **Agentic AI** – Autonomous decision-making and personalized career guidance

Novelty and Uniqueness

The **Career Counseling Companion** combines **Agentic AI, personalization, and intelligent career planning** into a single smart platform. Unlike traditional career counseling, it provides personalized, data-driven guidance tailored to each student's goals, skills, and interests.

Agentic AI Guidance – Acts as an intelligent digital career mentor that understands student needs and provides proactive recommendations.

Personalized Career Matching – Recommends the most suitable career paths based on academic background, skills, and interests.

Skill Gap Analysis – Identifies missing skills and suggests areas for improvement.

Learning Roadmap Generation – Creates customized learning plans with relevant courses, certifications, and projects.

Industry-Aware Recommendations – Aligns career suggestions with current industry trends and future opportunities.

IBM-Powered Solution – Built using **IBM watsonx Orchestrate**, **IBM watsonx.ai**, and **IBM Granite Models** for scalable and intelligent career guidance.

Git Hub Link

· Please create public github repository (Enable Add README button while creating repository)
with format –

Please test and Paste the working Github URL –<https://github.com/Tannu-Khandelwal/career-counselling-companion>

Make sure you have Uploaded following three files into your github repository

1) app.json file , 2) yourproblemstatement.pdf file 3) your projectpresntation.pptx file 4)any
other files

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Future Scope

1. AI-Powered Resume & Interview Preparation –

The system can automatically generate professional resumes, identify improvement areas, and conduct mock interviews with personalized feedback to improve employability.

2. Real-Time Job Market & Skill Analysis –

By analyzing current industry trends and job demands, the Career Counseling Companion can recommend in-demand skills, certifications, and career opportunities.

3. Personalized Learning & Certification Pathways –

The platform can suggest courses, internships, and certification programs based on a student's interests, strengths, and career goals.

Thank You!

Thank you for your time and interest.